Beyond Black–Scholes

Local volatility, stochastic volatility, and stochastic rates

Lecture 12

The model must explain the prices we observe

The question

Black–Scholes is a powerful benchmark. Which of its assumptions fail in the market, and what model extension addresses each failure?

1. Read the implied-volatility surface as evidence against constant volatility.
2. Build local volatility to fit today's vanilla-option surface.
3. Make volatility random to describe its future dynamics.
4. Make discounting random when the yield curve is a source of risk.

NO-ARBITRAGE The central trade-off is always the same: fit, realism, hedging, and tractability cannot all be maximized at once.

Why Black–Scholes is not enough

One constant volatility cannot fit a whole option market

Black–Scholes assumes, under $\mathbb{Q}$,

$$\frac{\mathrm{d}S_t}{S_t} = r\,\mathrm{d}t + \sigma\,\mathrm{d}W_t, \qquad \sigma \text{ constant.}$$

If the model were exact, Black–Scholes implied volatility would be the same for every strike K and maturity T.

What markets show

Implied volatility varies by strike: smiles and, in equities, a strong downside skew.

What markets also show

Implied volatility varies by maturity; uncertainty has its own term structure.

WATCH OUT A single number σ cannot reproduce a two-dimensional surface $(K, T) \mapsto \sigma_{\mathrm{imp}}(K, T)$.

The equity skew has an economic interpretation

For a fixed maturity, a typical equity surface has

$$\sigma_{\text{imp}}(K_{\text{low}}) > \sigma_{\text{imp}}(K_{\text{ATM}}) > \sigma_{\text{imp}}(K_{\text{high}}).$$

Why low-strike puts are expensive

They protect against severe equity declines. Strong demand for this insurance raises their market prices and therefore their implied volatilities.

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Answer

No. It is the Black–Scholes parameter that matches a market price.

Three extensions answer three different questions

Observed issue	Model extension	Main objective
Static smile across K, T	Local volatility	Fit vanilla prices today
Volatility moves through time	Stochastic volatility	Model smile dynamics and volatility risk
Discounting and curve move	Stochastic rates	Model bonds and the term structure

NO-ARBITRAGE These models do not replace no-arbitrage pricing. They replace restrictive assumptions about what is deterministic or constant.

Local volatility fits today's surface

Local volatility makes the diffusion coefficient state dependent

Under the risk-neutral measure,

$$\boxed{\frac{\mathrm{d}S_t}{S_t} = r\,\mathrm{d}t + \sigma_{\mathrm{loc}}(t, S_t)\,\mathrm{d}W_t.}$$

What remains

The model is one-factor and Markovian. Given t and S_t, the future law is determined.

What changes

Instantaneous variance is no longer constant:

$$\mathrm{Var}\left(\frac{\mathrm{d}S_t}{S_t}\middle|\mathcal{F}_t\right) = \sigma_{\mathrm{loc}}(t, S_t)^2\,\mathrm{d}t.$$

NO-ARBITRAGE Black–Scholes is the special case $\sigma_{\mathrm{loc}}(t, s) \equiv \sigma$.

The pricing PDE changes in one precise place

For a European payoff $g(S_T)$, the price $u(t,s)$ solves

$$u_t + rs\,u_s + \tfrac{1}{2}\sigma_{\mathrm{loc}}(t,s)^2 s^2 u_{ss} - ru = 0, \qquad u(T,s) = g(s).$$

Compare with Black–Scholes

The pricing logic, risk-neutral drift, and terminal condition are unchanged. Only the constant variance σ^2 becomes the local quantity $\sigma_{\mathrm{loc}}(t,s)^2$.

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Answer

No. It is a deterministic function of t, S_t, so the model remains one-dimensional.

Dupire extracts local volatility from call prices

Let $C(T, K)$ be the market price of a call by maturity and strike. With constant rates and no dividends,

$$\boxed{\sigma_{\text{loc}}(T, K)^2 = \frac{C_T(T, K) + rK\, C_K(T, K)}{\frac{1}{2}K^2 C_{KK}(T, K)}.}$$

Denominator

$$C_{KK}(T, K)$$

is the curvature in strike; it is linked to the risk-neutral density.

Numerator

$$C_T + rKC_K$$

captures how the call surface changes with maturity and strike.

Call convexity is a no-arbitrage input to Dupire

For an arbitrage-free call-price surface,

$$C_{KK}(T, K) \geq 0.$$

Formally,

$$C_{KK}(T, K) = e^{-rT} p^{\mathbb{Q}}_{S_T}(K),$$

where $p^{\mathbb{Q}}_{S_T}$ is the risk-neutral density.

YOUR TURN Why is applying Dupire's formula directly to a sparse table of noisy market quotes dangerous?

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Answer

The formula differentiates the surface, especially twice in strike. Interpolation noise is amplified and can produce negative or unstable local variances unless the input surface is smoothed and arbitrage controlled.

Local volatility calibrates exactly, but only statically

A practical calibration pipeline is:

$$\text{quotes} \longrightarrow \text{arbitrage-free call or implied-vol surface} \longrightarrow \text{smooth derivatives} \longrightarrow \sigma_{\mathrm{loc}}(T, K).$$

Strength

With a clean surface, local volatility can reproduce today's European vanilla prices across the quoted grid.

Limitation

It prescribes future smile motion through a deterministic function of spot and time; those dynamics are often not realistic.

Stochastic volatility models volatility itself

A random variance factor creates new dynamics

A generic stochastic-volatility model has

$$\frac{\mathrm{d}S_t}{S_t} = r\,\mathrm{d}t + \sqrt{V_t}\,\mathrm{d}W_t^{(1)},$$

$$\mathrm{d}V_t = b(t, V_t)\,\mathrm{d}t + a(t, V_t)\,\mathrm{d}W_t^{(2)}, \qquad \mathrm{d}\langle W^{(1)}, W^{(2)}\rangle_t = \rho\,\mathrm{d}t.$$

The state is now (S_t, V_t)

Even if two paths have the same spot S_t, their option values can differ when their current variance levels V_t differ.

NO-ARBITRAGE The pair (S, V) is Markovian, but the stock alone is generally not.

Heston adds mean reversion, vol-of-vol, and correlation

A standard Heston specification is

$$\frac{\mathrm{d}S_t}{S_t} = r\,\mathrm{d}t + \sqrt{V_t}\,\mathrm{d}W_t^{(1)}, \qquad \mathrm{d}V_t = \kappa(\theta - V_t)\,\mathrm{d}t + \eta\sqrt{V_t}\,\mathrm{d}W_t^{(2)}.$$

Parameter	Economic role
θ	long-run variance level
κ	speed of mean reversion toward θ
η	volatility of volatility
ρ	correlation between spot and variance shocks

Negative spot-volatility correlation generates an equity skew

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Answer

$$\boxed{\rho < 0.}$$

A negative stock shock tends to be accompanied by a positive variance shock. Low-strike puts become relatively valuable, generating a downward implied- volatility skew.

NO-ARBITRAGE Random variance creates smile curvature; correlation controls much of its asymmetry.

Mean reversion gives a useful variance forecast

From

$$\mathrm{d}V_t = \kappa(\theta - V_t)\,\mathrm{d}t + \eta\sqrt{V_t}\,\mathrm{d}W_t,$$

the conditional mean is

$$\boxed{\mathbb{E}[V_T \mid V_t = v] = \theta + (v - \theta)e^{-\kappa(T-t)}.}$$

YOUR TURN Suppose $v > \theta$. Is the expected future variance above or below today's variance, and why?

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Answer

Below v, because the drift $\kappa(\theta - v)$ is negative. The larger κ, the faster the expected pull toward the long-run level.

More realism makes hedging and pricing harder

Pricing

The option value is $u(t, s, v)$, so the PDE has two state variables. Characteristic-function methods, PDEs, and Monte Carlo are commonly used.

Hedging

There are two sources of risk but often only the stock and cash are traded. The model is generally incomplete; variance risk cannot be perfectly hedged with the stock alone.

NO-ARBITRAGE In practice, liquid vanilla options are often used alongside the stock to manage volatility exposure.

Local and stochastic volatility solve different problems

	Local volatility	Stochastic volatility
Volatility	deterministic $\sigma_{\text{loc}}(t, S)$	random factor V_t
Main strength	exact static vanilla fit	realistic smile dynamics
Dimension	one factor	at least two factors
Hedging	closer to complete diffusion hedging	generally incomplete

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Answer

Stochastic volatility: it retains a separate random variance state and is designed to describe its future evolution.

Stochastic rates make discounting risky

With stochastic rates, the discount factor is random

If the short rate is r_t, the pathwise discount factor is

$$D(t,T) = \exp\left(-\int_t^T r_s \, \mathrm{d}s\right).$$

The zero-coupon bond is its risk-neutral value:

$$\boxed{P(t,T) = \mathbb{E}^{\mathbb{Q}}[D(t,T) \mid \mathcal{F}_t].}$$

Key distinction

With constant rates, $P(t,T) = e^{-r(T-t)}$ is deterministic. With stochastic rates, discounting itself is a source of uncertainty.

Bond prices, zero rates, and forwards encode the same curve

For maturity $T > t$,

$$P(t,T) = e^{-R(t,T)(T-t)}, \qquad R(t,T) = -\frac{\log P(t,T)}{T-t},$$

and the instantaneous forward rate is

$$f(t,T) = -\partial_T \log P(t,T).$$

NO-ARBITRAGE The bond-price curve $T \mapsto P(t,T)$, zero-rate curve $R(t,T)$, and forward curve $f(t,T)$ are three parameterizations of the same term structure.

Worked curve example: extract a zero rate and a forward rate

Suppose

$$P(0,1) = 0.96, \qquad P(0,2) = 0.90.$$

The two-year continuously compounded zero rate is

$$R(0,2) = -\frac{1}{2}\log(0.90) = 5.27\%.$$

The simply compounded one-year forward rate for $[1,2]$ is

$$L(0;1,2) = \frac{P(0,1)/P(0,2) - 1}{2-1} = \frac{0.96}{0.90} - 1 = 6.67\%.$$

Reading the numbers

The forward rate exceeds the two-year average zero rate: the curve embeds higher implied rates over the second year than over the first.

Coupon bonds are portfolios of zero-coupon bonds

A bond paying coupons c_i at dates T_i, and principal F at T_n, is worth

$$\boxed{B_t = \sum_{i=1}^{n} c_i P(t, T_i) + FP(t, T_n).}$$

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Answer

$$B_0 = 4(0.96) + 104(0.90) = 97.44.$$

The entire pricing problem reduces to discount factors at the cash-flow dates.

Short-rate models compress the curve into a few factors

A short-rate model starts from a process r_t, then derives bond prices:

$$P(t,T) = \mathbb{E}^{\mathbb{Q}}\left[\exp\left(-\int_t^T r_s \,\mathrm{d}s\right)\middle|\mathcal{F}_t\right].$$

Vasicek

$$\mathrm{d}r_t = a(b - r_t)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t.$$

Gaussian and mean reverting, but rates can become negative.

CIR

$$\mathrm{d}r_t = a(b - r_t)\,\mathrm{d}t + \sigma\sqrt{r_t}\,\mathrm{d}W_t.$$

Mean reverting with a square-root diffusion that supports positivity.

HJM models the forward curve directly

Instead of modeling only r_t, Heath–Jarrow–Morton specifies

$$\mathrm{d}f(t,T) = \alpha(t,T)\,\mathrm{d}t + \sigma(t,T)\,\mathrm{d}W_t.$$

The no-arbitrage restriction

Under the risk-neutral measure, the drift $\alpha(t,T)$ is not freely chosen. Once forward-rate volatility is specified, no-arbitrage determines the drift.

Strength

It starts from the object the market quotes: the whole forward curve.

Cost

Without additional structure, the model is infinite-dimensional.

When must rates be modelled stochastically?

Often a reasonable approximation	Stochastic-rate model matters
Short-dated vanilla equity option	Bond or bond option
Small rate sensitivity	Caplet, floorlet, or swaption
Short horizon	Long-dated equity or hybrid payoff
Deterministic discounting acceptable	Discounting is itself a key risk

YOUR TURN Which column fits a five-year callable bond: left or right?

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Answer

Right. Its cash flows and exercise decision are directly exposed to the evolution of the yield curve.

Calibration and model choice

Calibration targets depend on the model class

Model class	What is fitted	Typical challenge
Local volatility	vanilla implied-vol surface	smooth, arbitrage-free derivatives
Stochastic volatility	surface through $\kappa, \theta, \eta, \rho, V_0$	non-linear optimization and stability
Stochastic rates	initial curve plus caps, swaptions, bond options	fit curve and dynamics together

NO-ARBITRAGE Calibration is not merely parameter estimation. It aligns a model with tradable market prices before the model is used for valuation or risk.

Model choice is a product decision, not a beauty contest

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Answer

Usually not. The product is exposed both to volatility dynamics and to rate risk. A hybrid framework, or at least explicit stochastic-rate modelling, may be necessary.

Practical rule

Use the least complex model that captures the material risk factors and can be calibrated reliably to liquid hedging instruments.

The three extensions form one modeling hierarchy

market prices $\longrightarrow$ model assumptions $\longrightarrow$ calibration $\longrightarrow$ pricing and hedging.

1. Local volatility fits the current vanilla surface.
2. Stochastic volatility adds a random volatility state and richer dynamics.
3. Stochastic rates add a random discount curve and fixed-income risk.
4. More realism comes with more state variables, calibration work, and hedging uncertainty.

Final check: start from the risk you need to model

YOUR TURN A desk is hedging a barrier option and needs to reproduce today's full vanilla smile surface accurately. Which model is the natural first benchmark, and what caveat should remain in mind?

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YOUR TURN A desk is hedging a barrier option and needs to reproduce today's full vanilla smile surface accurately. Which model is the natural first benchmark, and what caveat should remain in mind?

Answer

Local volatility is the natural first benchmark because it fits today's vanilla surface. Its future smile dynamics may nevertheless be unrealistic, so the hedge can be model dependent.

A richer model earns its place only by explaining a material risk.